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like fuel economy, speed and driving performance. These are analysed to take corrective actions.

Technical specifications

FarEye comes packaged in an enclosure measuring a reasonable 8×3×1.2 cm³ and weighing 150 gm. It uses an 8V to 28V supply for its operation and needs 1 mA in sleep mode and 90 mA during transit. It operates at an optimal temperature of 40-65°C and relative humidity of 80 per cent. The GPS channel operates at L1 frequency in coarse/acquisition code. It has a hot starting time of less than one second, tracking sensitivity of 160 dBm and horizontal position accuracy of 5 metres. The GSM client is RoHS-compliant, lead-free and soldered on an ENIG multi-layered PCB.

Obstacles

Kumar had to face challenges in order to get the product up and running. He shares, "There were issues in making the system easy to use for parents. They aren't the most technology-savvy people out there and we had to keep that in mind while finalising the design. The new design is very easy to follow with a detailed write-up in case of further help. The new user interface is capable of drawing up reports, which are designed to help the school authorities get all the requisite details in just one click. To make matters transparent, we even provided an application programming interface (API) to schools where parents can see the bus details without logging in."

Features

FarEye is programmed to bring a live view of the selected vehicle on a map, which is constantly updated throughout the duration of the route. It is capable of pan-India coverage. The application allows you to geo-fence, i.e., you can define an area on the map so that you automatically get an SMS alert on your phone when the vehicle enters that zone.

FarEye is designed to generate reports incorporating a host of different



FarEye bus-tracking device

critical parameters as per the users' demands. Some of the parameters are speed, number of stoppages, idling time, distance travelled along with a one-year history of the routes taken. The application also stores/sends details of sudden brakes applied (if any) and presents them in a report for the extra-paranoid parents. There is a mobile website (www.fareye.in) to assist in tracking the vehicle details via a smartphone. A host of similar features are covered completely in the documentation.

Future plans

When asked about plans for monetising this solution, Kumar replied, "Our plan to monetise FarEye is pretty basic. We provide the solution to schools and charge them for the system along with a per SMS charge. It is, as of now, a finished product. But, we are ready to take feedback from the parents, schools and the industry to add features to it as and when required. The present system took us close to a year to complete. This is including the pilot and all the prototypes required to reach the final product. We have already completed the provisional patents and are in the process of filing a complete patent."

As of now, parents can heave a sigh of relief as their kids can now be safer and more in their sights than ever before. ●

The author is a tech correspondent at Efy Bengaluru